

Inhibition of the Progression of Type 2 Diabetes in the C57BL/6J Mouse Model by an Anti-diabetes Herbal Formula

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The effects of a specifically prepared anti-diabetic herbal formula (ADHF) on the course of established diet-induced type 2 diabetes in animal subjects has been studied. In a C57BL/6J mouse model for diet induced type 2 diabetes, intervention for 12 weeks using ADHF as a diet supplement resulted in a significant inhibition of diabetes related changes in major organs usually targeted by type 2 diabetes and a significant reduction in circulating levels of glucose and insulin. Young male mice were randomly assigned to receive *ad libitum* exposure to either a standard rodent chow diet or to a high fat, high simple sugar, low fibre diet (diabetes induction diet), respectively for 8 weeks. All mice fed the induction diet developed abnormally high blood glucose levels at 8 weeks. Animals with confirmed diet induced diabetic blood glucose levels were again randomly assigned into one of three groups (10 subjects per group), one group was thereafter fed only the diabetes induction diet and the other two groups were thereafter fed the diabetes induction diet into which ADHF had been mixed at 4% or at 8% final concentrations. Normal mice were also randomized into two groups that were fed either a regular diet alone or 8% ADHF mixed in the regular diet. Blood glucose levels markedly increased over the 20 weeks of study in the diabetic mice fed the diabetes induction diet only. In contrast, diabetic mice fed induction diet into which 4% or 8% ADHF had been incorporated showed significantly decreased blood glucose and insulin levels over the time of the study. Additional parameters significantly reduced in diabetic mice fed ADHF included insulin resistance and histopathological changes in the pancreas and liver. This is the first report to our knowledge to show *in vivo* evidence for significantly decreased circulating glucose and insulin levels and a significant reduction of progressive damage to major target organs by the addition of an herbal diet supplement to a diabetes induction diet proven to be capable of causing and maintaining type 2 diabetes. Copyright © 2003 John Wiley & Sons, Ltd.

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INTRODUCTION

Among the sugar imbalance conditions frequently detected in human populations, one of the most prominent is non-insulin dependent diabetes mellitus (NIDDM) or type 2 diabetes, which is a disease frequently encountered in obese adolescents and adults (Harris *et al.*, 1987). During the progression of type 2 diabetes, the development of both insulin resistance and α - and β -cell dysfunctions appear to be the basic metabolic abnormalities leading to the long-term disease (Saad *et al.*, 1991). Despite advances in understanding the aetiology and treatments of such sugar imbalance problems with their possible linkage to genetics, diet, exercise, autoimmunity and viral aetiology factors (de Fronzo *et al.*, 1992; Knowler *et al.*, 1995), Western medicine has had limited success in maintaining normal sugar balance during long term treatment of persons with chronic type 2 diabetes (Sartor *et al.*, 1980; Keen *et al.*, 1982). In

contrast, herbal diet supplements and medicines from the Far East have been used for centuries to strengthen elements of the body's resistance to sugar imbalances (Keij, 1981; Teng, 1983; Huo *et al.*, 1988; Pang, 1991). The proprietary herbal diet supplement (ADHF) used in our studies was prepared from herbs which have been well studied and widely used in China and the Pacific Rim (Yu *et al.*, 1994; Zhang and Huo, 1994; Huo *et al.*, 1995) to help correct sugar imbalance problems.

To test whether or not ADHF could alter the progressive course of increasing sugar imbalances associated with type 2 diabetes, mice were selected of the C57BL/6J strain, well recognized for developing obesity, hyperglycaemia and NIDDM on a reproducible time scale and schedule (Surwit *et al.*, 1988, 1991). C57BL/6J mice carry a genetic disposition for developing significant sugar imbalances within 2 months if given *ad libitum* access to a high fat, high simple carbohydrate and low fibre diet (Surwit *et al.*, 1988; 1991). Sugar imbalance syndromes similar to those of type 2 have been identified in several animal species (Reaven, 1991), but until recently there has been no satisfactory small animal model of this disease available for the testing of herbal and traditional diet supplements for potential beneficial effects.

In this study we have investigated the induction of abnormally high sugar imbalances in special genetically

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susceptible mice by feeding them an induction diet with a high fat, high simple sugar and low fibre content after which we have restored the sugar balance in these same mice by continuing to feed them the induction diet into which had been incorporated a special herbal diet supplement (ADHF).

MATERIALS AND METHODS

Herbal materials. Anti-diabetes herbal formula (ADHF), a generous gift from PhytoCell Research, Inc. (San Antonio, Texas USA), was composed of traditional Chinese herbs dried and as water extracts, which had been selected, prepared, activated and mixed according to proprietary formulation ingredients and methods (US Patent 6093403; Huo *et al.*, 2000) under GMP conditions. The powder mixture included eight major herbal components, i.e. Ginseng Radix 17%, Rehmannia Radix 17% g, Astragali Radix 10%, Trichosanthis Radix 10%, Ophiopogonis Radix 10%, Puerariae Radix 10%, Lycii Fructus 10% and Discoreae Rhizoma 10%. ADHF powder at 4% and 8% final concentrations was mixed with other ingredients of the diabetes induction diets when prepared by the commercial manufacturer (Bio Serve, Youngstown, NJ).

Animal subjects. C57BL/6J male mice, especially bred to develop diet induced significant sugar imbalances reflective of type 2 diabetes (Surwit *et al.*, 1988), were purchased at 3 weeks of age from Jackson Laboratories (Bar Harbor, ME) through Laboratory Animal Resources, UTHSC-SA. All animal work was performed in accordance with approved protocols by Institutional Animal Care and Use Committee (UTHSC-SA). The C57BL/6J male mice were acclimatized for 1 week before random assignment into study groups. The animals were housed at five per cage in a temperature and atmosphere controlled room with 12 h light-dark cycles. Ten animals were used in each experimental (treatment) and control group in repeated (two) trials under the same protocol. No replacement animals were used in any group as a result of death or illness during these studies.

Regular diet (RD). This diet, used to feed the normal non-diabetic control group, consisted of standard rodent chow containing 23% protein, 4.5% fat, 6.0% fibre, 8.0% ash and 56% complex carbohydrate (Purina, St Louis, MO). ADHF mixed to a final concentration of 8% with RD was fed to a normal mouse control group to identify any effects of the ADHF in normal subjects (RD + 8% ADHF). These regular diets and water were fed *ad libitum* to animals in the normal control groups at all times in each trial.

Diabetes induction diet (DID). DID was a specially prepared high fat, high simple carbohydrate and low fibre diet (Diet F1850, lot 20811, Bio-Serv, Frenchtown, NJ). The mineral content of DID was similar to that of the RD. The diabetes induction diet into which had been mixed ADHF at 4% final concentrations was labelled 'DID + 4% ADHF' (Diet F3335, lot 20800, Bio-Serv, Frenchtown, NJ), while ADHF at an 8% final concentration in DID was labelled 'DID + 8%

ADHF' (Diet F3226, lot 20801, Bio-Serv, Frenchtown, NJ). The percentages of fat, fibre and simple sugars (disaccharides) were markedly different between the RD and DID, while small differences in the percentages of these three ingredients were detected between DID alone and the two treatment diets, i.e. DID + ADHF mixtures at final concentrations of 4% and 8%. The special diabetes induction and treatment diets and water were fed *ad libitum* to animals in the experimental diabetes induction groups in each trial.

Consumption profile. The amount of food consumed daily per cage was measured by weighing the remaining food, then adding a measured amount (g) of new food to animals in each of the control and treatment diet groups. The quantity of food eaten by subjects on each type of diet was studied.

Blood glucose and serum insulin. Blood samples were collected by retro-orbital sinus puncture into small capillary tubes. Approximately 50 µL of a fresh blood sample was placed on each of duplicate test strips and the blood glucose content was read in a validated One Touch Basic glucose measurement system (Lifescan Inc). The glucose measurement system values were validated by 10 serial measurements of the manufacturer's standard reference high and low glucose control solutions.

Blood insulin levels were determined by collection of blood samples drawn from tail vein punctures into 200 µL capillary tubes, which were centrifuged at 4 °C, 1500 × g for 10 min. Insulin levels were determined for each separated serum according to protocols described by the manufacturer of ¹²⁵I-RIA insulin kits (Intar. Corp.) Intra- and inter-assay coefficients of variation for the insulin determinations were 8.5% and 4.2%, respectively.

Body weight. Each animal in each group was weighted weekly with an electrobalance (Sartorius, Model B310S).

Index of insulin resistance. The relative index of insulin resistance was calculated using relationships between the body weight (BW), diet (amount consumed (g/mouse/day)), blood glucose (BG), and serum insulin (SI) levels according to the following formula:

$$\text{Index of insulin resistance (IIR)} = \frac{\text{BW} \times \text{BG} \times \text{SI}}{\text{Diet} \times 1000}$$

Histopathological and immunocytochemical examination. At the end of the experimental course (6 months) the mice were killed by transcardial perfusion of 4% paraformaldehyde containing 15% of picric acid for 12 min or by decapitation. The major organs in perfusion-fixed animals routinely removed included heart, pancreas, kidney, brain, lung, adrenal gland, spleen, thymus, lymph nodes (popliteal fossa), sciatic nerve, testis and pituitary gland. The same major organs in decapitated animals were also quickly removed, directly frozen in liquid nitrogen and stored for cryostat sectioning. All paraformaldehyde-fixed tissues were embedded in paraffin, sections 6 µm thick cut and then stained with haematoxylin and eosin. Frozen liver tissue was cut 12 µm thick with a cryostat microtome and stained with oil red O. All sections were examined under a light microscope. Pancreas sections were processed for immunohistochemistry with several primary antibodies,

including polyclonal anti-insulin and anti-glucagon (DAKO). Immunostaining was done according to the avidin-biotin peroxidase complex (ABC) method (Hsu *et al.*, 1981; Yao *et al.*, 1994) using biotinylated secondary antibodies and according to instructions in kits obtained from Vector Laboratories (Burlingame, CA). Double immunolabelling for anti-insulin and anti-glucagon was performed as previously described by Kloppel *et al.* (1983) and Yao *et al.* (1995).

Histochemical methods, employed for demonstrating the fine changes of the glomerular basement membrane of kidney and of islets of Langerhans of the pancreas, included periodic acid-Schiff's (PAS) reaction and amyloid staining as previously described by Yao and coworkers (1995). Measurements of the severity of fatty degeneration of livers and the diameters of islet of Langerhans in the pancreas body portions were made using Bioquant OS/2 software for microscopic image analysis to determine the density of lipid deposits in the hepatic cells in oil red O stained liver sections and in immunostained pancreatic tissues (Stefan *et al.*, 1982).

Schedule of experiments. After adaptation, mice (4 weeks old) were randomized into groups of 10 each. A normal non-diabetes healthy control group was fed RD only. Another normal control group was fed RD plus 8% ADHF. The diabetes type 2 positive control group was fed DID only. The two treatment groups of subjects were fed either DID + 4% ADHF or DID + 8% ADHF, respectively beginning at confirmation of diabetes establishment (12 weeks). All animals were monitored daily for the onset of polyuria and additional bedding changes were made as appropriate.

Body weights were determined weekly. At biweekly intervals blood specimens were collected from mice in each of the control and treatment diet groups. Blood glucose and serum insulin assays were performed and the index of insulin resistance calculated. At the completion of the studies at 6 months, blood samples were collected from all mice in the control and treatment diet groups, after which they were killed and appropriate tissues for pathological evaluations were immediately collected and processed.

Statistical analysis. Repeated trials (two total) were performed using identical protocols with time-response curves for major event markers generated with individual data points presented as the arithmetic mean $\pm$ SD. The statistical significance of the results was evaluated by the Student's *t*-test (one- and two-way analysis of variance) and probability values of less than 0.05 were considered to be significant.

RESULTS

Development of diet induced type 2 diabetes in the C57BL/6J mouse model

The special diabetes induction diet (DID) was found to be completely effective in stimulating the onset of diet induced type 2 diabetes in C57BL/6J mice on an extremely reliable schedule, i.e. within 8 weeks. The content percentages of fat, fibre and simple sugars, i.e. disaccharides and types of carbohydrates were markedly

and significantly different from those contained in the regular diet (RD), although the protein, ash and moisture contents were similar (data not shown).

Mice in the diabetes positive (DID only) group developed increasingly heavier body weights and increasingly higher blood glucose over the 6 month studies (Figs 2 and 3; Table 1). Eight weeks after being fed the induction diet (DID only), blood glucose levels in all mice consistently indicated the establishment of type 2 diabetes. At the end of the studies at 6 months, mice in the diabetes positive (DID only) groups showed markedly high levels of circulating insulin and blood glucose, markedly heavy body weights, very high index of insulin resistance and diabetes-related pathological changes in major organs, including the pancreas and liver, compared with what was detected in diabetes negative mice fed RD (Table 1). None of the diets used in the study, i.e. regular or induction alone or mixed with ADHF, caused any side effects, toxicities, adverse effects or deaths when fed for over 6 months to the mice.

Food consumption profiles

During the induction of diabetes in the mice fed DID only, from weeks 4 to 12, the quantity of food eaten was similar. However, by the end of the studies, the amounts of food consumed were significantly lower in the diabetes positive groups fed DID only than in those treatment groups fed DID + 4% or 8% ADHF (Fig. 1). The addition of ADHF to RD had no apparent effect on quantity of food eaten in the normal control groups (Fig. 1).

Body weight, blood glucose and serum insulin levels

Body weights of normal control animals in the RD and RD + 8% ADHF groups increased as the subjects grew during the 6 month studies (Fig. 2). In contrast, the body weights of animals fed DID only (diabetes positive controls) showed large increases which were significantly greater than those detected in animals fed either of the two treatment diets (Fig. 2; Table 1). Likewise,

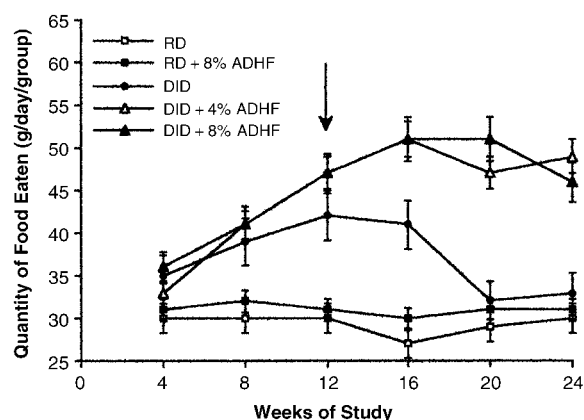


Figure 1. Quantity of food consumed by both normal non-diabetic animal groups was similar (RD and RD + 8% ADHF), as was that consumed by the two treatment groups (DID + 4% ADHF and DID + 8% ADHF). In contrast, food consumed by the diabetic positive control group (DID only) decreased significantly over the 12 week post induction period. Arrow shows start time of treatment by feeding ADHF in DID. Values represent mean $\pm$ SD.

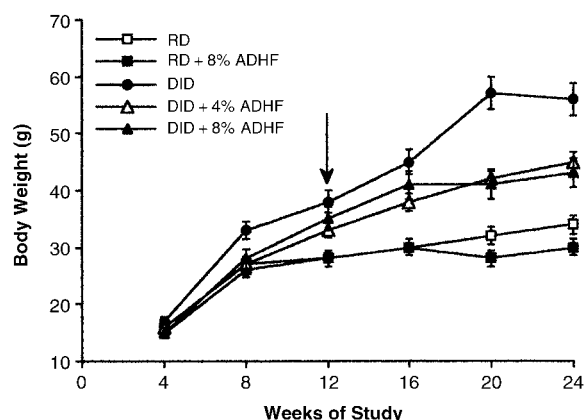


Figure 2. Body weights of normal non-diabetic control animals (RD and RD + 8% ADHF) slowly increased over the 6 month study as did those of mice in the two treatment groups (DID + 4% ADHF or DID + 8% ADHF). In contrast, the body weight of diabetic positive control (DID only) mice showed rapid and large increases in weight. Arrow shows start time of treatment by feeding ADHF in DID. Values represent mean $\pm$ SD.

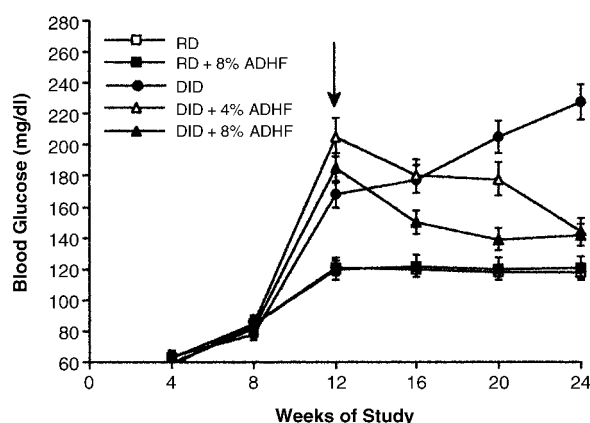


Figure 3. Blood glucose levels measured at 4 week intervals in mice from all groups showed slow increases in the normal non-diabetic mice (RD and RD + 8% ADHF). Marked increases over the study course were detected in mice in the diabetic positive control group (DID only) while mice in the treatment groups (DID + 4% ADHF and DID + 8% ADHF) had significantly reduced blood glucose levels over the 12 week post induction treatment period. Arrow shows start time of treatment by feeding ADHF in DID. Values represent mean $\pm$ SD.

blood glucose levels slowly increased in normal control group animals, while marked, vigorous increases were detected in animals fed DID alone during the induction of diabetes times, i.e. 4–12 weeks (Fig. 3). Thereafter (12–24 weeks), animals on the DID + ADHF treatment diets showed significant decreases in blood glucose levels compared with the levels detected for the diabetes positive control group fed DID only (Fig. 3; Table 1). As further seen in Table 1, serum insulin levels at 6 months were markedly increased in the DID only (diabetes positive control) group compared with levels detected in each of the two ADHF fed treatment groups. Index of insulin resistance for animals in each group at 6 months showed up to a 10 times higher value in the DID only (diabetes positive) group, while animals on the treatment diets (DID mixed with 4% or 8% ADHF) had levels of insulin resistance similar to those in normal RD controls (Table 1).

Diabetes-related pathological changes in some major organs

Gross examinations showed that mice in the diabetes positive control groups (fed DID only) were the heaviest

and most obese with severe fatty livers and great amounts of fat tissue deposited in the peritoneal cavity and in subcutaneous tissues. In normal control (fed RD) group mice, normal weights, no lipophilia, no fatty livers and normal amounts of deposited fat were detected. Animals that received a diet of DID + 4% and 8% ADHF during the treatment times of 12–24 weeks showed body weights lower than the disease controls, little increased amounts of deposited fats and not severe fatty livers.

Histopathological evaluations of endocrinal pancreas tissues and cells revealed in the DID only mice markedly larger islets of Langerhans at 6 months than in the normal (RD) control and treatment (DID + ADHF) groups (Figs 4 and 5). Quantitative evaluations of glucagon-positive A cells and insulin-positive B cells in the islets from all subjects found significantly greater numbers of both α and β cells in the pancreatic tissues from diabetic positive (DID only) subjects (Figs 4 and 5). The islet sizes and A cell numbers in samples from the DID + ADHF treatment group mice were similar to those detected in the non-diabetic normal controls.

Histopathological examination of liver sections showed no adipose infiltration in hepatic cells from RD

Table 1. Effect of DID mixed with 4% and 8% ADHF on serum insulin, blood glucose, body weights and index of insulin resistance in C57BL/6J mice after 6 months^a

Group ^b	Insulin (μ U/mL)	Blood glucose (mg%)	Body weight (g)	Index IR ^c
RD	27.4 $\pm$ 2	107.9 $\pm$ 4	31.3 $\pm$ 1	86.6
DID	183.6 $\pm$ 15	228.5 $\pm$ 5	55.4 $\pm$ 3	1144.6
DID + 4% ADHF	41.1 $\pm$ 3 ^{de}	143.2 $\pm$ 5 ^{de}	43.4 $\pm$ 2 ^{de}	128.3 ^{de}
DID + 8% ADHF	46.4 $\pm$ 5 ^{de}	142.3 $\pm$ 6 ^{de}	44.9 $\pm$ 1 ^{de}	116.8 ^{de}

^a Values represent mean $\pm$ SD of 10 animals in each group.

^b RD, regular diet of standard Purina rodent chow; DID, diabetes induction diet; DID + ADHF, anti-diabetic herbal formula as percent in final mixture of DID. [Each diet containing DID was prepared by Bio-Serve, Frenchtown, NJ].

^c Index IR, Index of insulin resistance.

^d Difference in relation to diabetes positive control (DID) group.

^e $p < 0.01$.

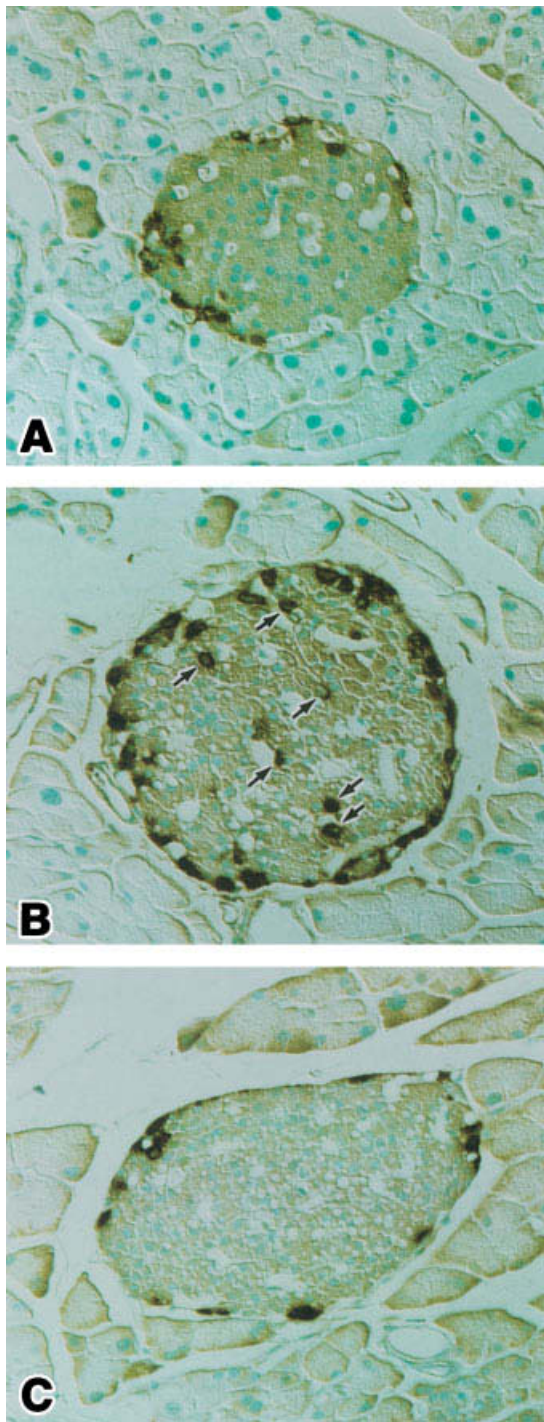


Figure 4. Photomicrographs of glycogen immunoreactivity showing A cell numbers and function status. Arrow shows dark stained A cells within islet section. Paraffin sections ABC method, methylgreen counterstained X240, A: RD – normal nondiabetic control; B: DID only – diabetic positive control; C: DID + 8% ADHF – treatment group.

fed mice, but in the type 2 diabetes positive control group mice, all showed severe fatty infiltrations and fatty degeneration of liver cells (Fig. 6).

In contrast, liver sections from mice in both treatment groups (DID + ADHF 4% and 8%) showed liver cells as almost normal with very little hepatic fatty degeneration (Fig. 6). The distribution of small lipid droplets, their size and numbers and areas of degeneration were significantly smaller and fewer in ADHF

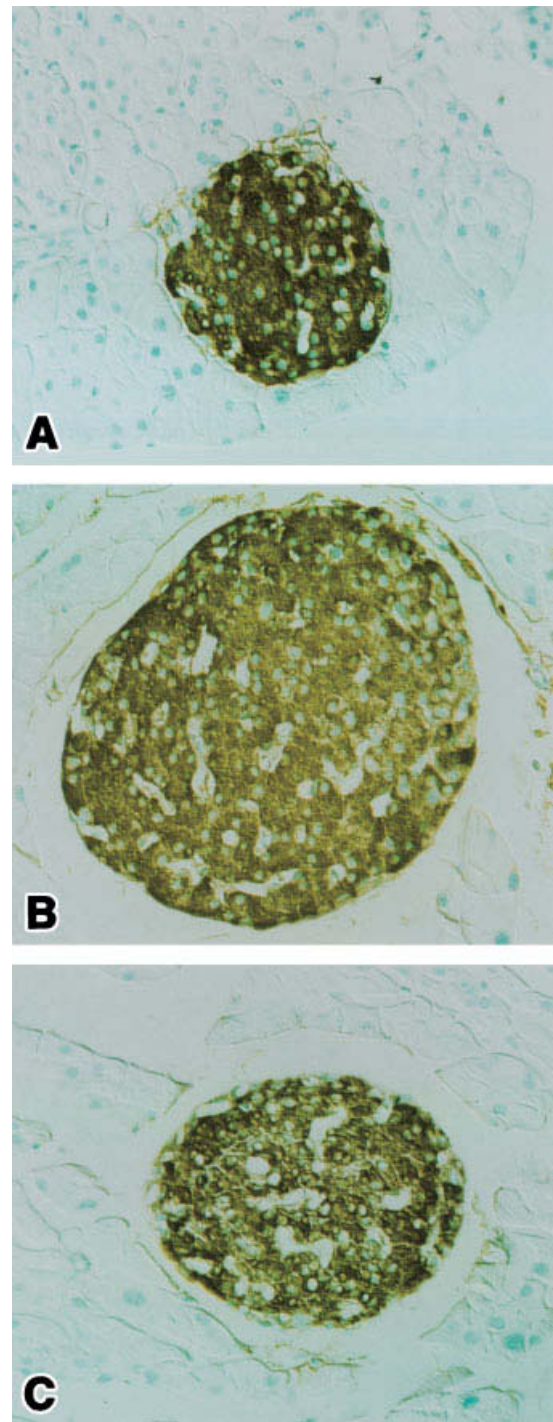


Figure 5. Photomicrographs of insulin immunoreactivity showing beta (β) cell numbers and function status in different groups. Paraffin sections, ABC method, methylgreen counterstained X240, A: RD – normal nondiabetic; B: DID only – diabetic positive control; C: DID + 8% ADHF – treatment group.

treated than those seen in DID only (diabetes positive) animals.

In mice from the different control and treatment groups, no obvious lipid deposits or arteriosclerosis were found in aorta intima and coronary arteries nor were obvious changes detected in cardiac muscle, brain, lungs and lymph organs (thymus, spleen, lymph nodes) at 6 months.

In the DID only (diabetes positive) group, moderate degeneration including cloudy swelling and vacuolar

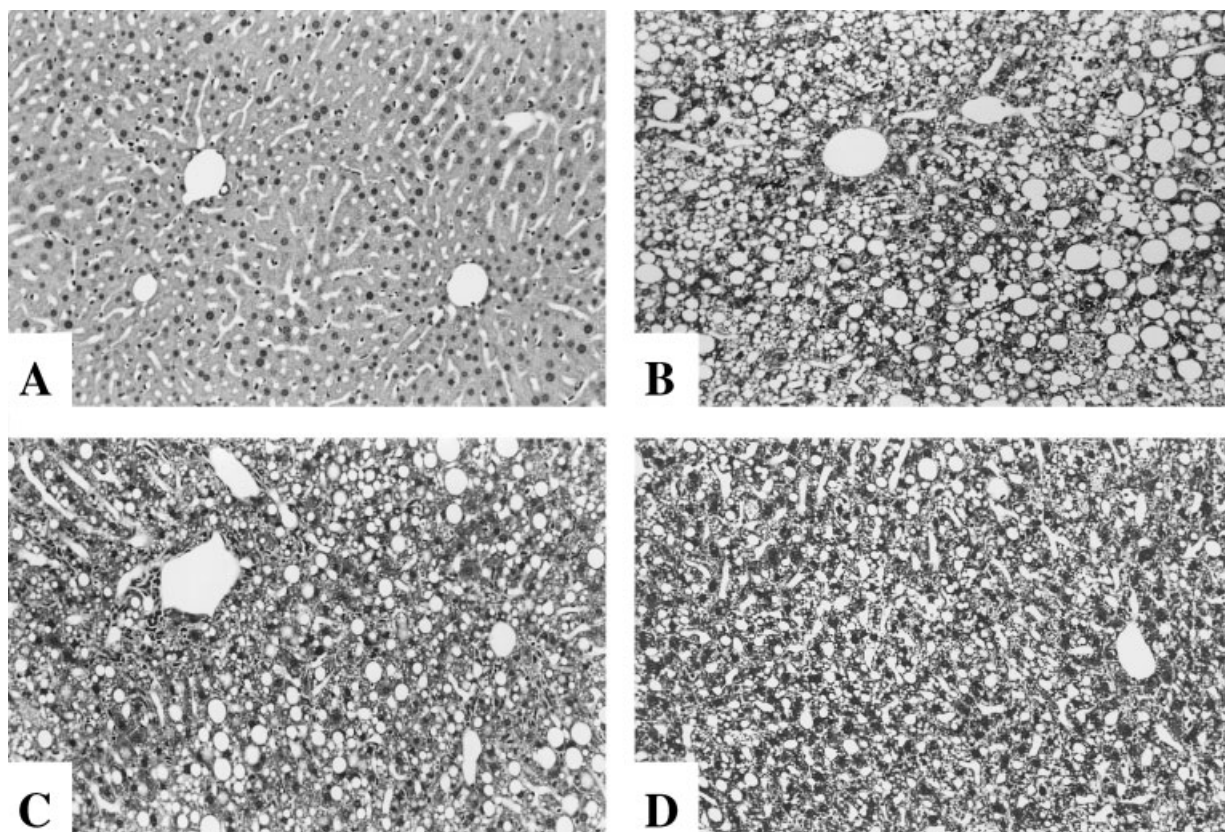


Figure 6. Photomicrographs of paraffin fixed liver sections from control and treatment mice. A: normal nondiabetic control (FD); B: diabetic positive control (DID only). Note most severe and significant fatty degeneration. C and D: treatment groups C (DID + 4% ADHF), D (DID + 8% ADHF) show markedly less extensive and fatty degeneration H and E $\times 200$.

degeneration of the convoluted tubules of the kidney were observed (data not shown). These changes were not detected in kidney tissues from mice in any other control and treatment diet groups.

DISCUSSION

The success of Surwit and co-workers (1988, 1991) in creating diet induced type 2 diabetes in C57BL/6J mice provided an opportunity for assessing the efficacy of a new dietary supplement formulated to reduce damage to sensitive organs caused by early stage type 2 diabetes. The results of the present studies confirmed the excellent and reliable diet induced type 2 diabetes model as described by Surwit and coworkers in 1988. This model uses genetically predisposed C57BL/6J mice which develop NIDDM-like symptoms on a regular, defined schedule when fed only a diabetes induction diet containing high fat, high simple carbohydrate and low fibre.

Previously Surwit and coworkers (1988, 1991) suggested that diet induced diabetes in C57BL/6J mice could depend on the development of obesity that follows prolonged access to the high fat, high simple sugar induction diet. Results from our present studies concur with this suggestion. In the diabetes positive controls, diet induced diabetes and obesity were coincident; however, intervention at the time of initial confirmation of the onset of type 2 diabetes in the mice by supplementing ADHF to the diet inhibited the subsequent development of obesity in these mice in spite of continued

prolonged access to the induction diet (Coleman, 1982; Kloppel *et al.*, 1983).

Surwit and co-workers (1988, 1991) reported that the C57BL/6J mouse model of type 2 diabetes appears to provide a better model of human type 2 diabetes than other *in vivo* models in which sugar imbalances are caused by genetic mutations and chemical or drug induced changes. Type 2 diabetes in humans is associated with genetic background, obesity, and a 'natural' induction diet usually high in calories, fat and simple carbohydrates with low fibre.

The C57BL/6J mouse type 2 diabetic model therefore follows closely NIDDM in humans since this model has a genetic basis, obesity associated with the diabetes, can be diet induced, and has many other symptoms seen in NIDDM of humans (Surwit *et al.*, 1988).

Rats or mice induced into diabetes by alloxan or streptozotocin have been widely used to study the effects of new drugs and herbal medicines as treatments for type 1 and type 2 diabetes in short-term studies (Okyar *et al.*, 2001). In addition, animal hereditary models, such as KK-Ay mice, which have hyperglycaemia automatically occurring with maturation, has allowed researchers to study type 2 diabetes in this mice model with Kampo medicines, unrelated to the diet consumed (Okumura *et al.*, 2001). We believe that the non-stressful induction of diabetes in the C57BL/6J mice provided us with a better model of diabetes induction and for evaluation of anti-diabetes treatments, especially those of the diet supplement type, than those which depend upon destruction of diabetes associated organs and tissues by drug and poison inducing agents.

Regarding the cell constituency of the pancreatic islets in type 2 diabetics, the numbers of A cells and the islet size are usually increased (Hoet *et al.*, 1995; Stefan *et al.*, 1982). Since islet sizes, low A cell numbers and B cells in islets from treated subjects on the DID + ADHF diets resembled those seen in the islets of normal controls in our studies, this appears to reflect a strong inhibiting effect of ADHF on the proliferation of A cells, which in turn helped reduce the secretions of glucagon and help prevent hyperglycaemia. In this same cascade of events making up a possible mechanism of action, it is reasonable to speculate that the body does not have to have high levels of insulin production, therefore, the insulin positive B cells can remain in normal numbers (Hoet *et al.*, 1995; Rahier *et al.*, 1983).

In the 12 weeks after the beginning of therapy, namely from the start time of addition of ADHF to the DID diet, ADHF supplemented mice maintained their level of food consumption with little weight gain (Figs 1 and 2) while those in the DID alone group developed type 2 diabetes-like syndromes, including hyperglycaemia, obesity, hyperinsulinaemia at a time when their food consumption was found to be significantly decreased (Figs 1 and 2). Therefore, all diabetic parameters were maintained in the diabetic positive control mice in spite of markedly decreased amounts of the DID consumed.

Relationships between onset of obesity, fat accumulation, hyperglycaemia, high fat-high sugar diets and genetic factors causing developments of diabetes are unclear (Rebuffle-Scrive *et al.*, 1993). Our results strongly suggest that interaction with an ADHF diet supplement can prevent further development of obesity, hyperglycaemia, hyperinsulinaemia and pathological changes in major target organs such as liver and pancreas.

Interventions that have been proposed and tested for the modification and treatment of NIDDM are weight reduction, exercise, dietary modification, and administration of oral drugs (DeFronzo *et al.*, 1992; Knowler *et al.*, 1995). The results of these present studies strongly suggest that intervention with an herbal diet supplement, e.g. ADHF, at the time of early detection of hyperglycaemia in humans may prove to be of value in conjunction with other treatments for altering the course of development of type 2 diabetes. Preliminary results from recently completed studies in diabetes patients have shown ADHF to complement the activities of specific antidiabetes drugs (data in preparation for publication, Winters, Huo, Ramirez, Yao, 2001).

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